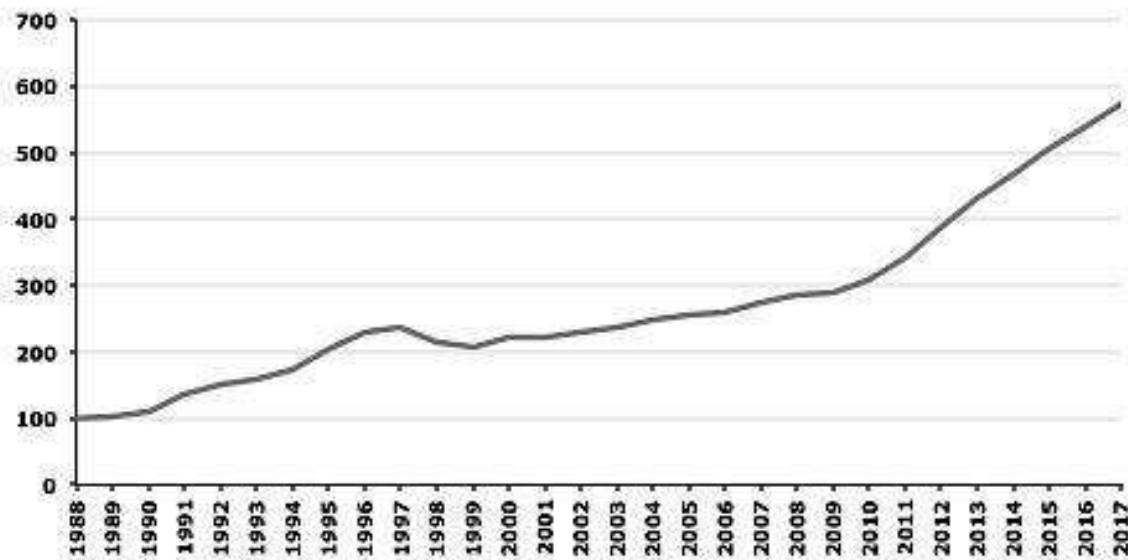


Office Works

- S&A Architects
- Employed
- 1 year after Bachelor - 2009
- 1 year after Masters - 2012

Malaysian All-House Price Index (MHPI), 1988-2017



Source: NAPIC (2018)



Design period: 2012

Jury Citation:

Pleasant residential estate with garden-like ambience affording generous outdoor living and the allure of indoor comfort.

PAM awards 2017

Multiple residential (Low rise) Silver

Project: Thistle Groove
Denai Alam

Developer: Sime Darby
Property

Architect: S&A Architect



Student work



Secondary market

According to Karim (2012) and Isnin et al. (2012), users, especially in mass housing, are not satisfied with their housing conditions. Most of them end up making alterations to their units before moving in by knocking down brick walls and building new ones to form rooms that suite their requirements (Rostam et al., 2012; Nurdalila 2012; Erdayu et al., 2010).



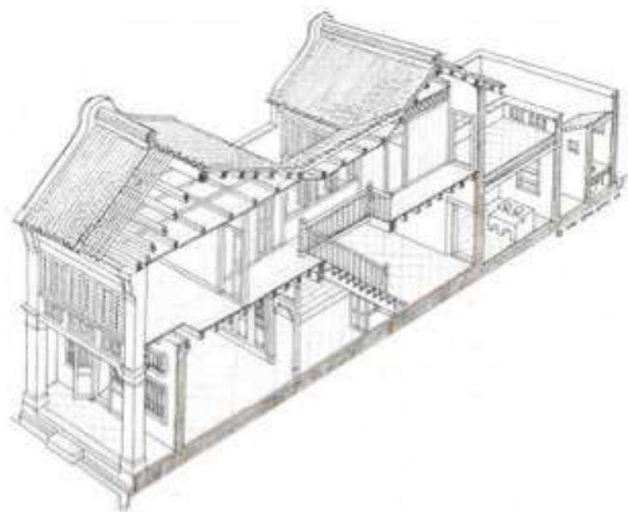
Figure 3. An example of an excessive alteration of a terraced house in Malaysia. On the right side is the original level of the terraced housing

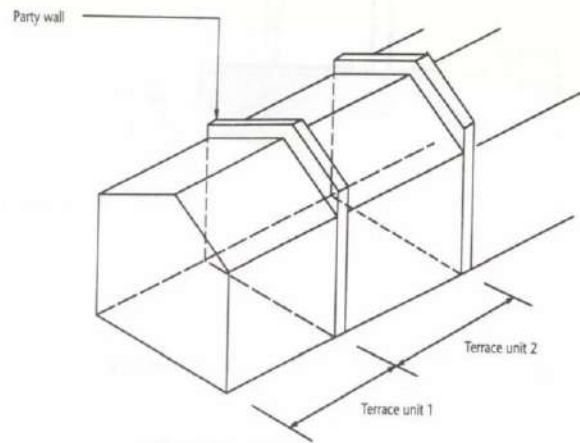
RENOVATION (REFORM)

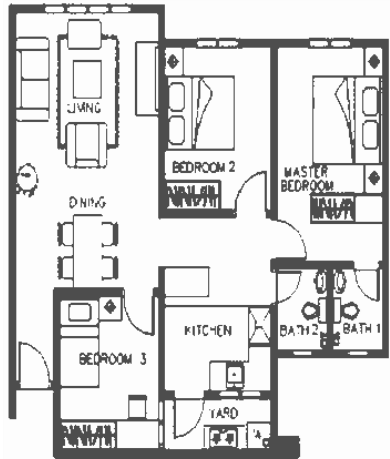


FIT-OUT









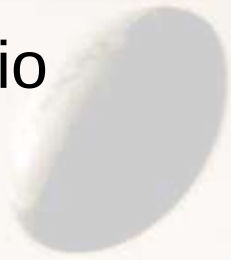


Lakar Design Studio

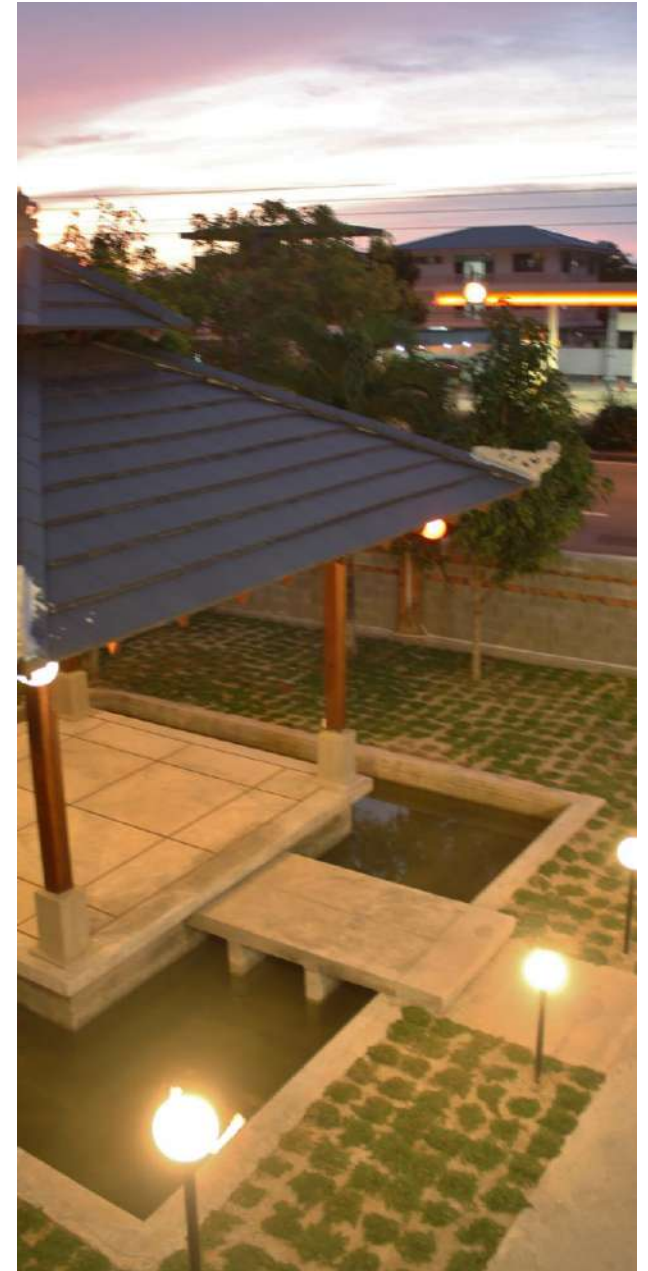
Back From Manchester, UK

Young designer with new ideas

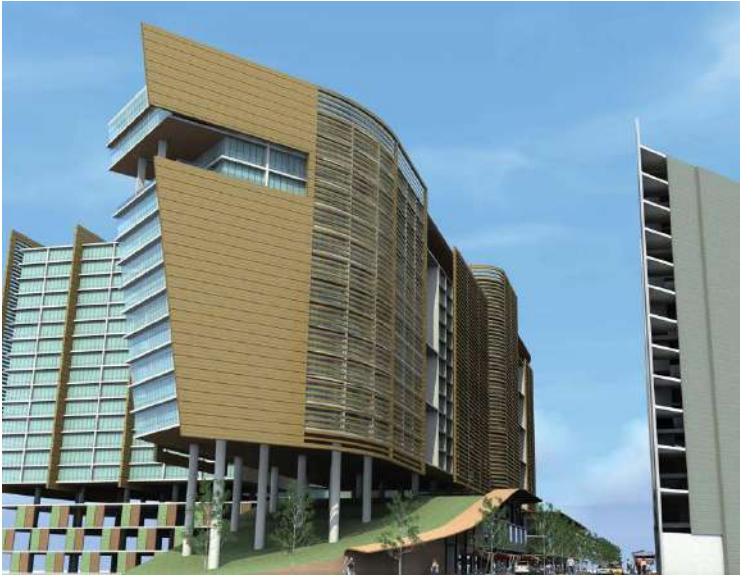
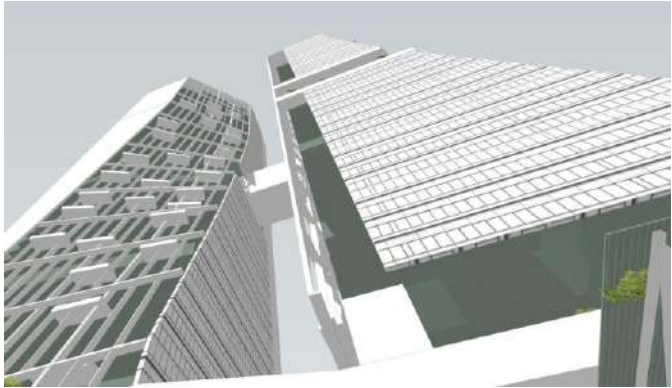
Exploring design freely

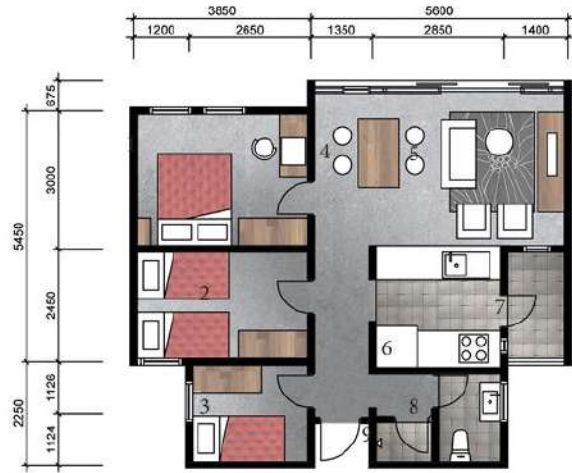
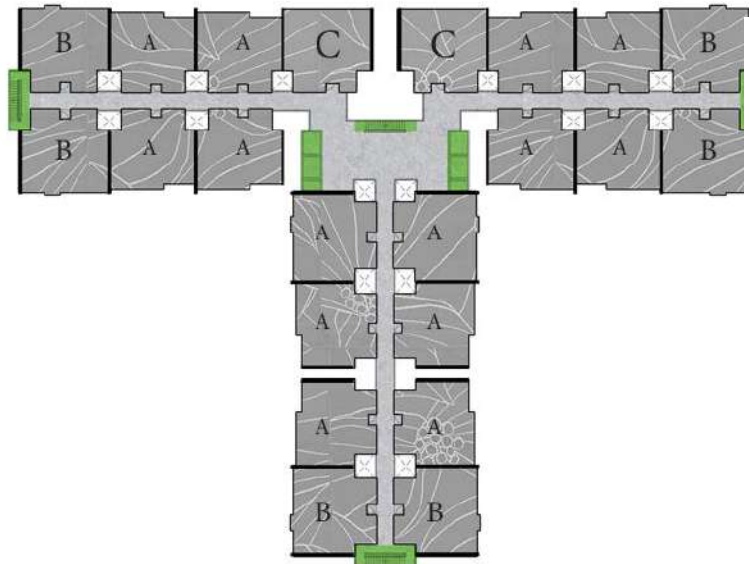


Sabah, Malaysia. Design & Build. Adaptive reuse







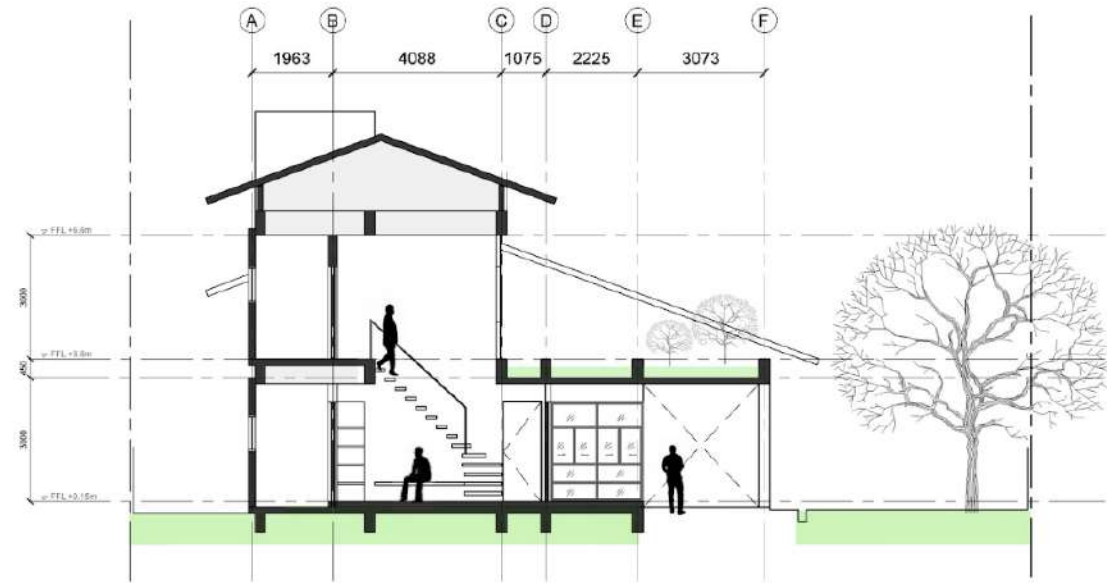
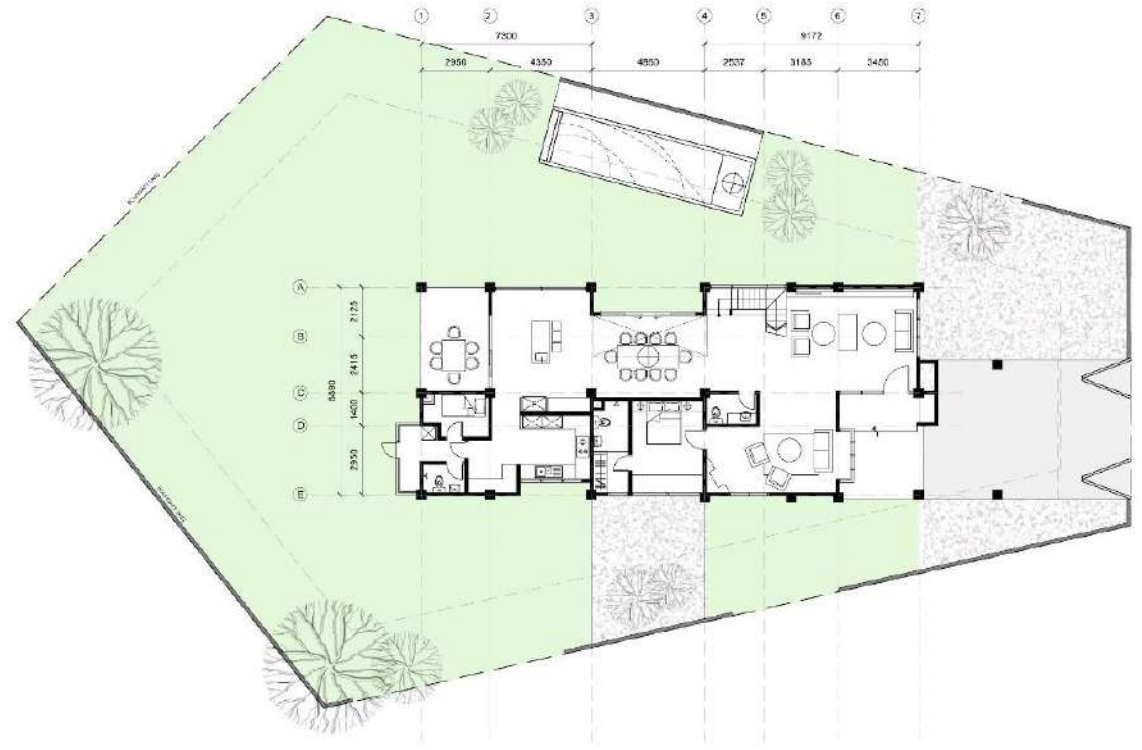


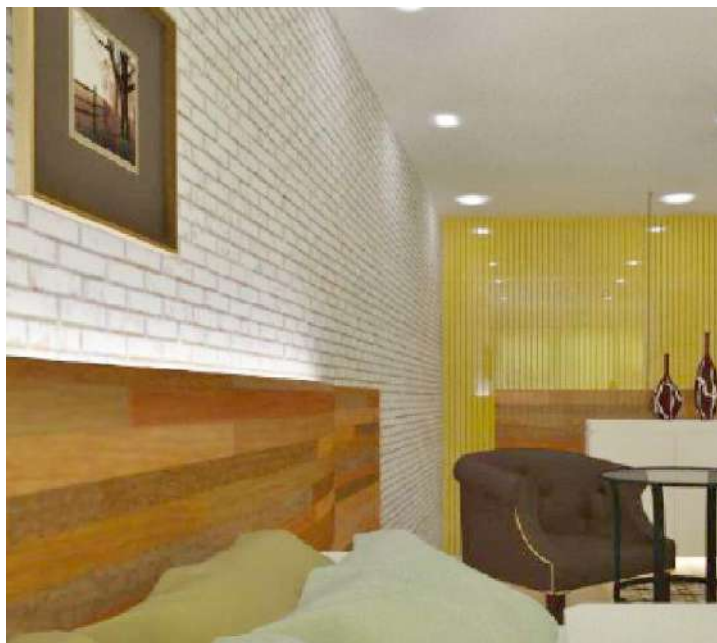
Deluxe unit 3r+1b TypeA 730 sqft













Lakar Design and Construction

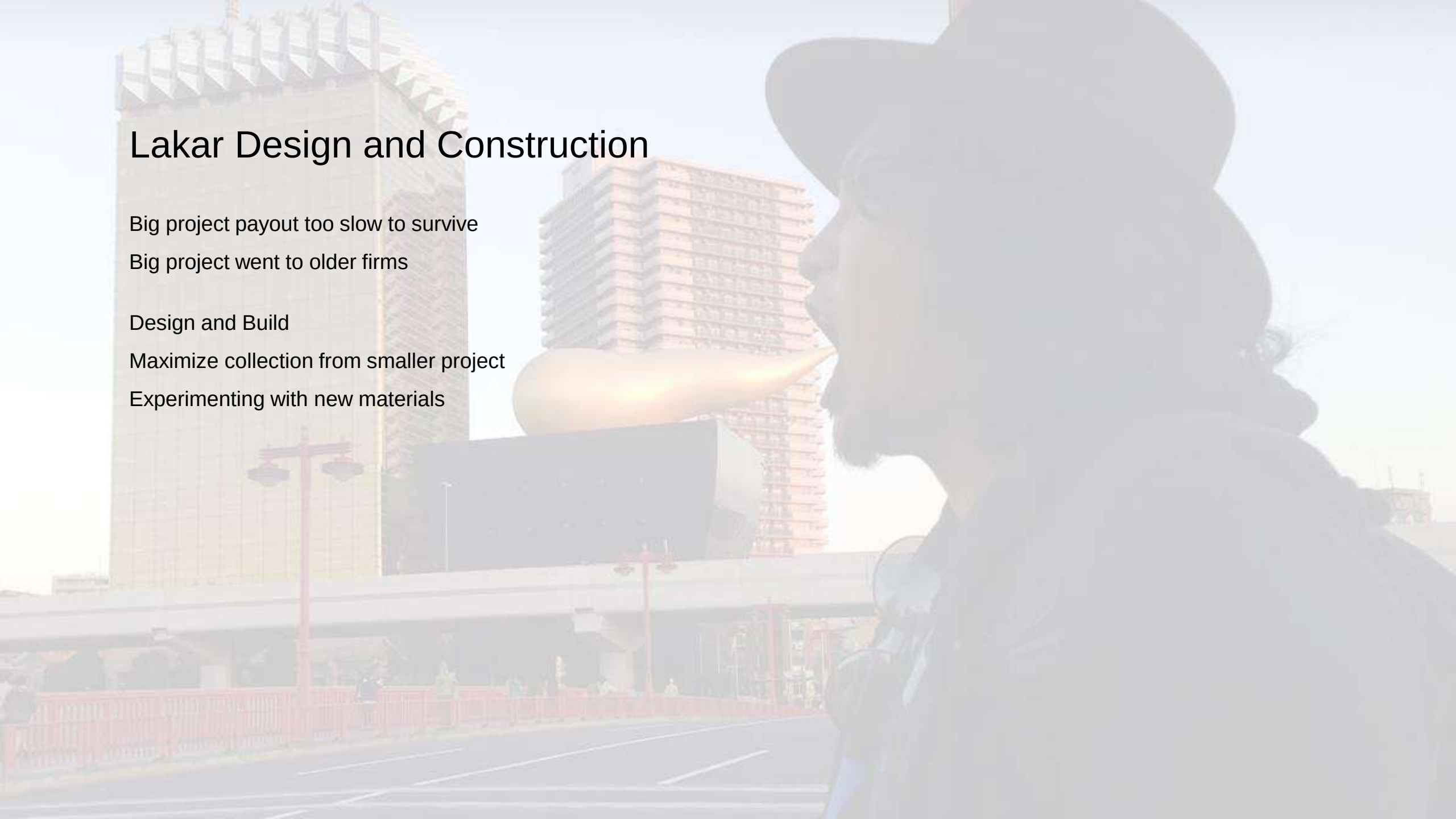
Big project payout too slow to survive

Big project went to older firms

Design and Build

Maximize collection from smaller project

Experimenting with new materials









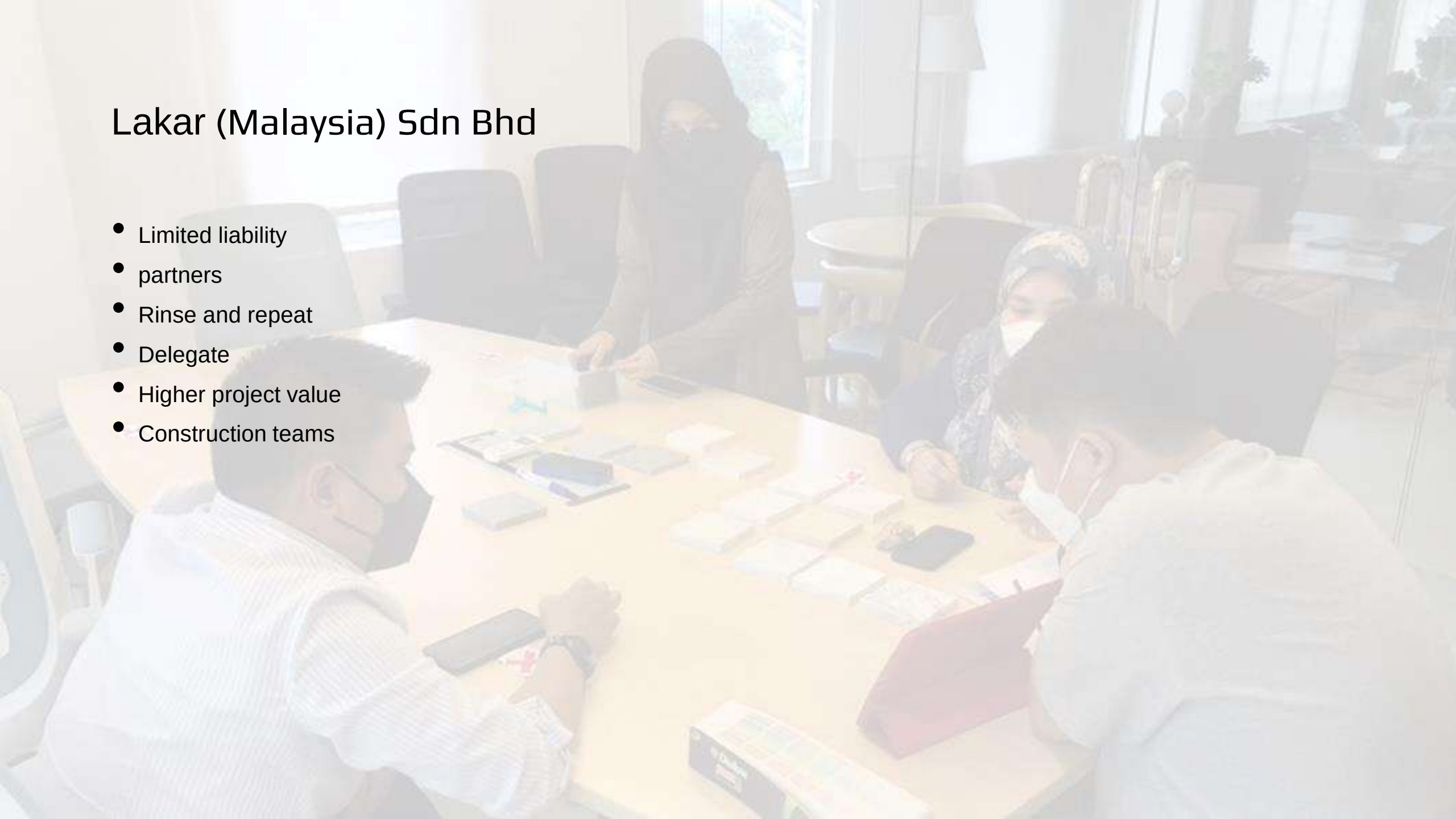


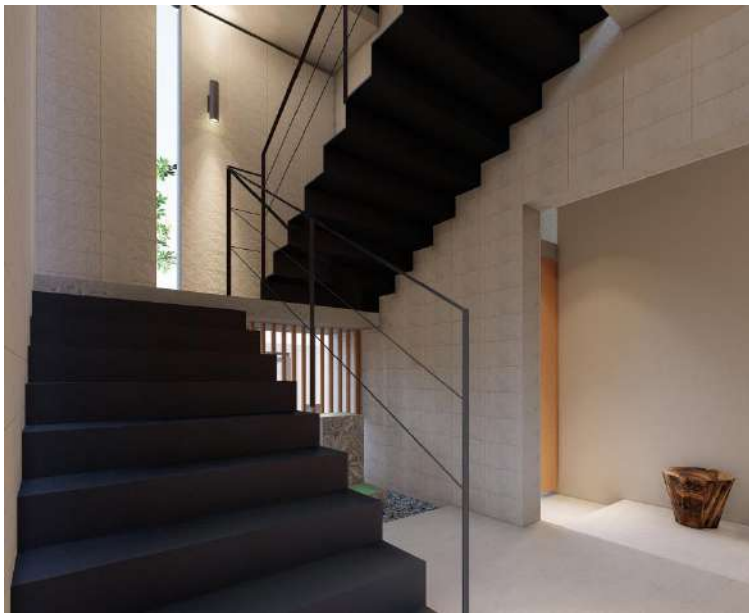




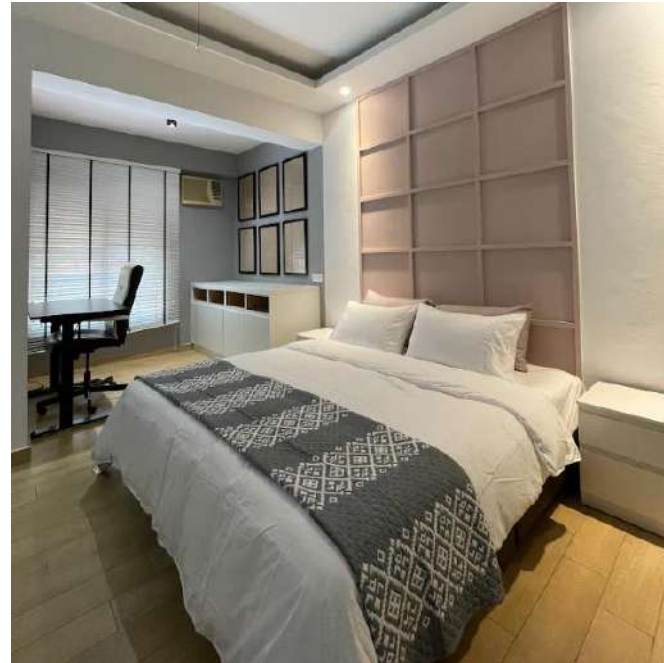
Lakar (Malaysia) Sdn Bhd

- Limited liability
- partners
- Rinse and repeat
- Delegate
- Higher project value
- Construction teams



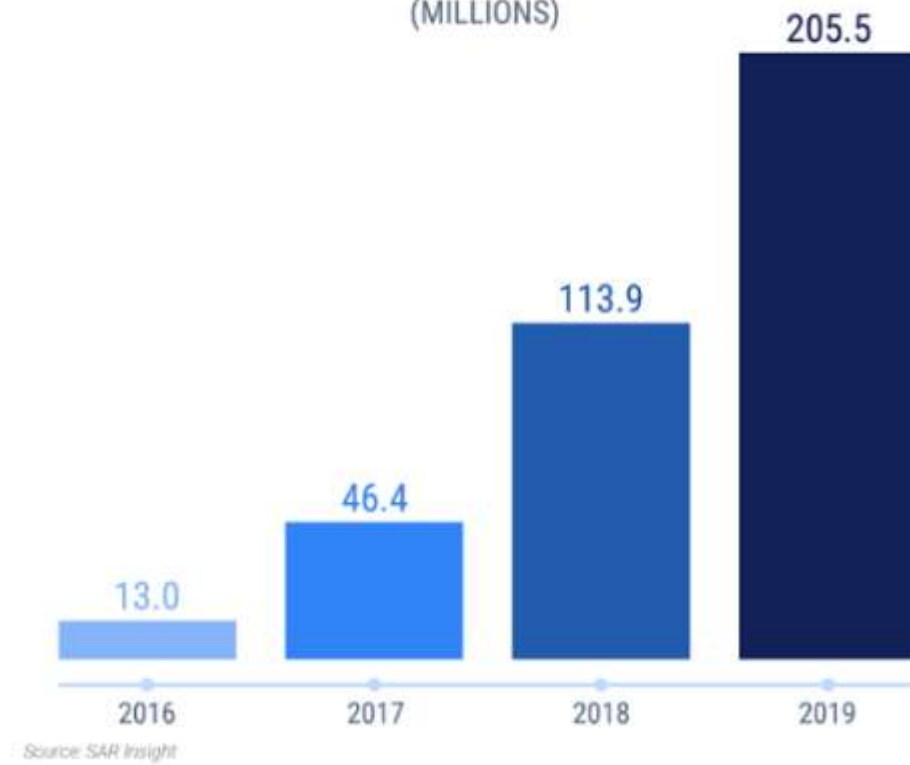






Smart Home Experiment

Worldwide Smart Speaker Shipments 2016 -19
(MILLIONS)

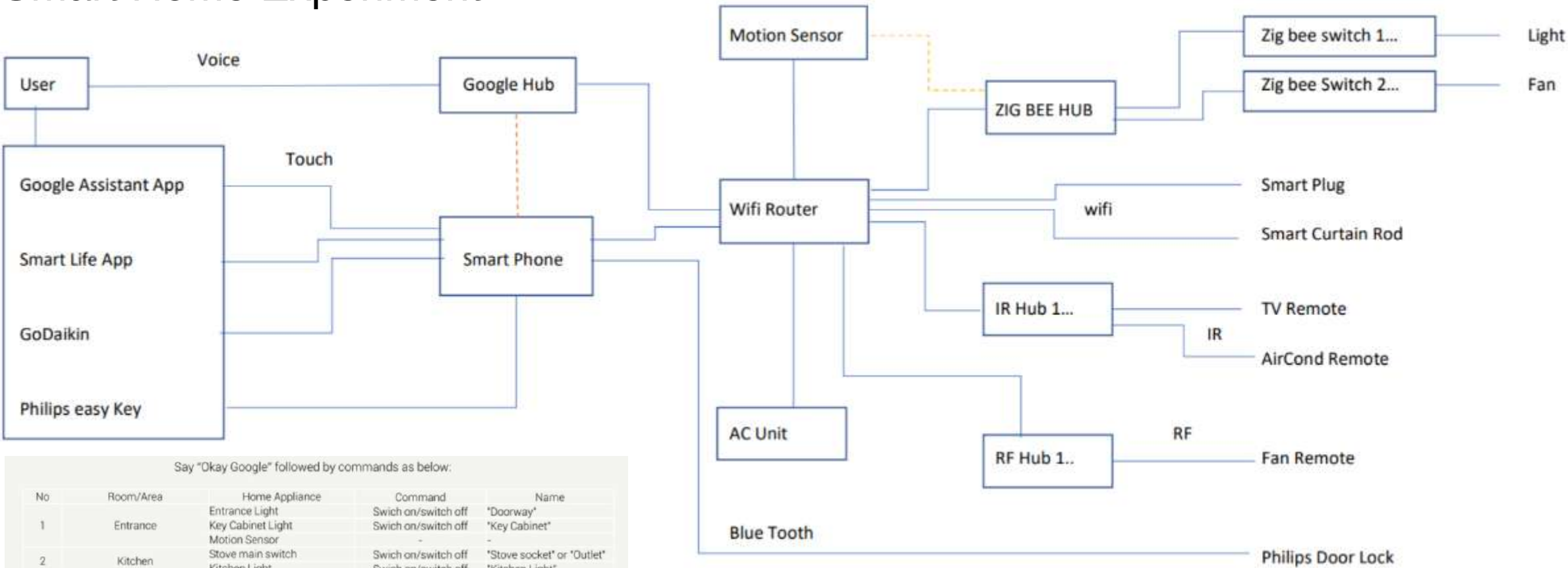


Smart Home Experiment





Smart Home Experiment



Say "Okay Google" followed by commands as below:

No	Room/Area	Home Appliance	Command	Name
1	Entrance	Entrance Light	Switch on/switch off	'Doorway'
		Key Cabinet Light	Switch on/switch off	'Key Cabinet'
		Motion Sensor	-	-
2	Kitchen	Stove main switch	Switch on/switch off	'Stove socket' or 'Outlet'
		Kitchen Light	Switch on/switch off	'Kitchen Light'
4	Dining Area	Dining hanging light	Switch on/switch off	'Dining Light'
		Smart TV		'TV' or 'Living TV'
		Model : LG 49UM7290PTD		
5	Living Area	Fan	Switch on/switch off	'Living Fan'
		Living light	Switch on/switch off	'Living Light'
		Curtain	Open / Close	'Living Curtain'
		Light behind TV panel	Switch on/switch off	'TV light'
		Corridor Light	Switch on/switch off	'Corridor light')
		Aircond	Switch on/switch off	'Living Air'
6	Bedroom 2	Light	Switch on/switch off	'Bedroom Light'
		Fan	Switch on/switch off	'Bedroom Fan'
		Curtain	Open / Close	'Bedroom Curtain'
		TV	Switch on/switch off	'TV' or 'Master TV'
7	Master Bedroom	Light	Switch on/switch off	'Master Light'
		Aircond	Switch on/switch off	'Master Aircond'
		Curtain	Open / Close	'Master Curtain'
		Fan	Switch on/switch off	'Master Fan'

Market adaption rate slow

- Cost
- Evolving technology
- Maintenance issue
- Spending Priority
- Cheap energy cost

Japan Research

- Pursue smart home development
- Energy efficient construction
- Prefabricated manufacturing
- Thermal insulation building
- Light weight building
- Quick construction turnaround
- Higher level of workmanship



BARRIERS TO ENERGY EFFICIENCY OF THE HOUSING STOCK,
Manchester school of architecture, united kingdom, ADAM RAMAN 2010,

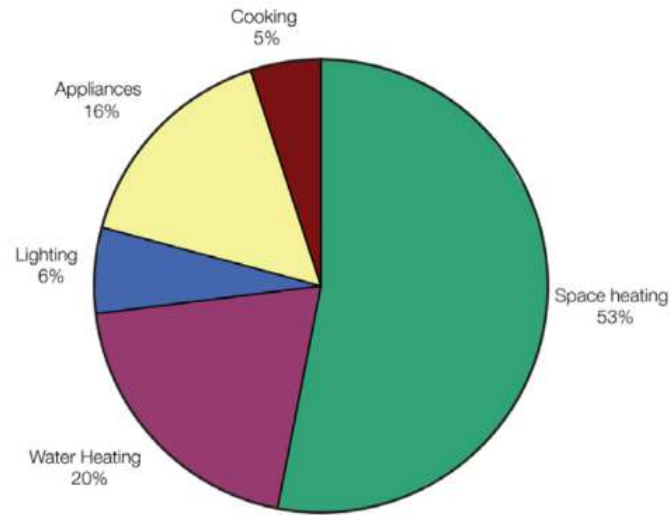
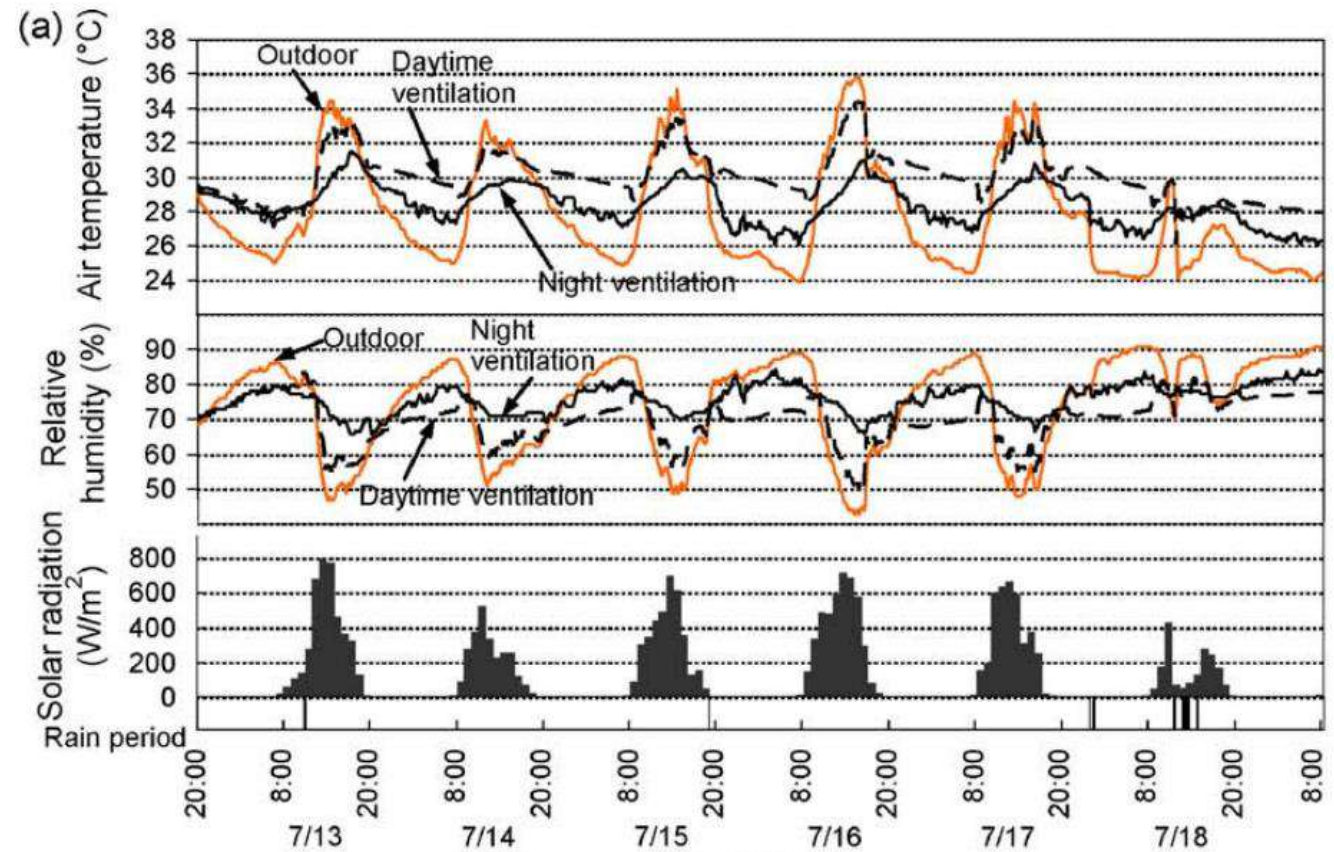


Figure 3 : Domestic Carbon Emissions – 2005, Source: (Department for communities and local government 2006)

- How to lower energy cost
- How to improve AC efficiency
- How to use AC sustainably



Solar Regenerated Daily Cycle Passive Dehumidifying Air-Conditioner: Storing Drying Potential in Solid Desiccant Through Daily Weather Cycles

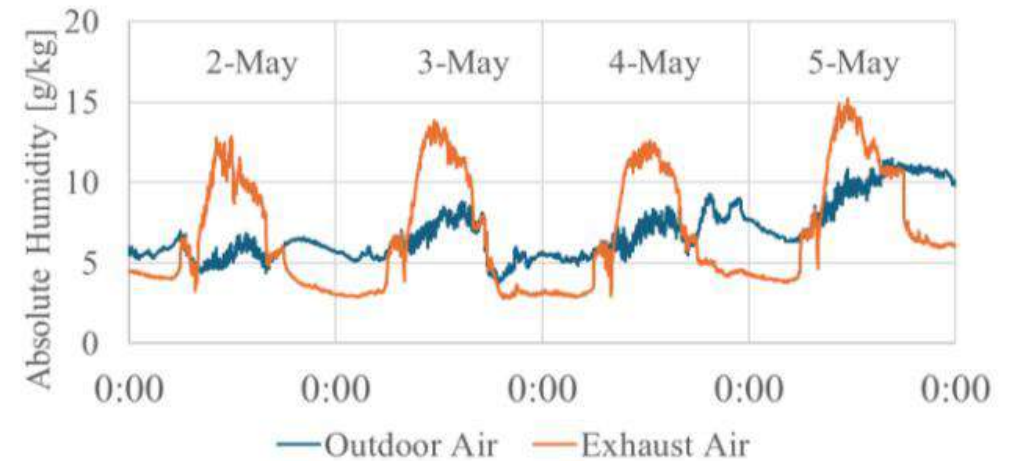
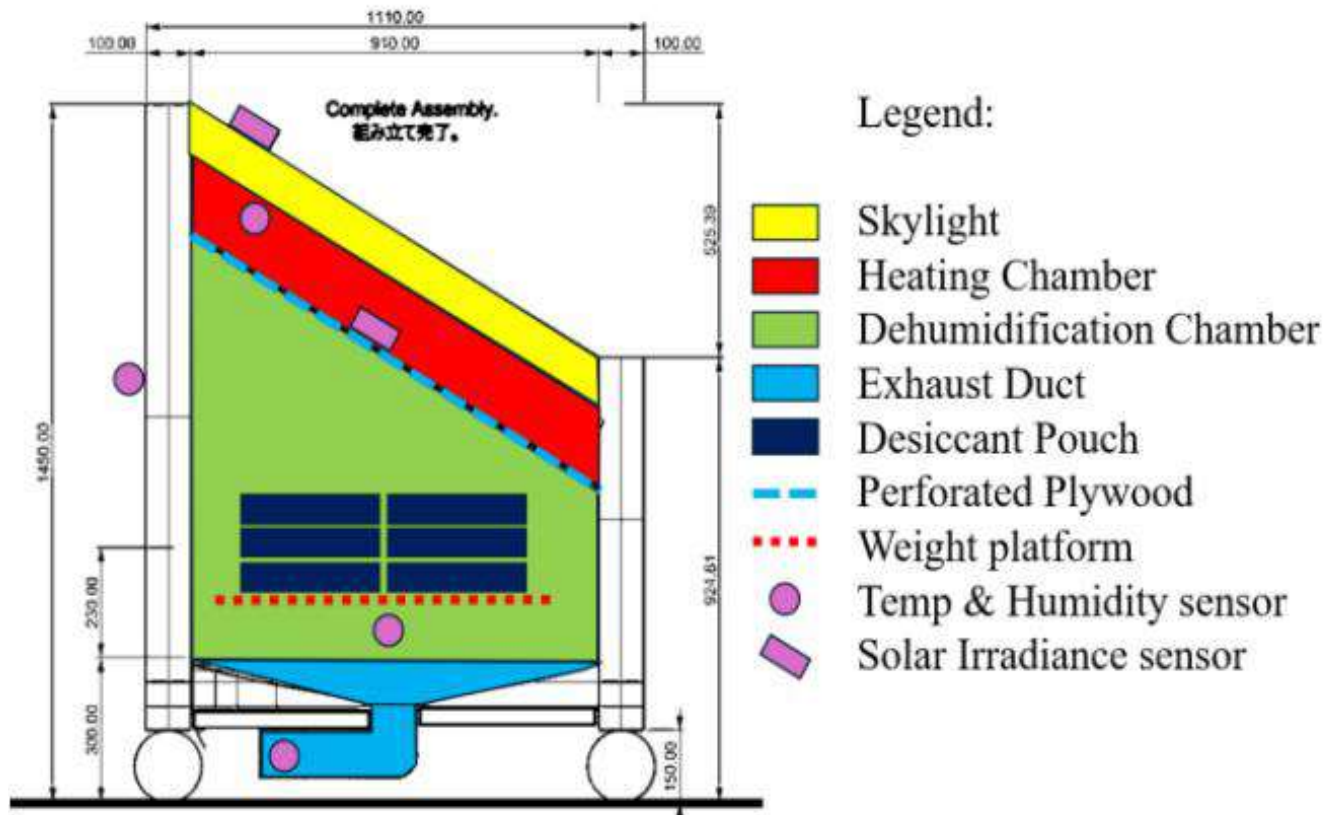


Fig. 7 Absolute Humidity over high solar irradiance days.

Future Goals

Living Buildings

Comfortable Buildings

Passive Buildings

